

On nominal event arguments: experimental evidence for a meaning-based perspective

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Abstract: In this paper, we present experimental insights on the semantics of English adjective-noun constructions such as *skillful dancer/ballerina/person*. We assume that adjectives like *skillful* are underspecified, and capture this by means of a context-dependent parameter in their semantics (Reinert 2024). We are interested in the (in)ability of nouns to provide a default interpretation for these adjective-noun constructions. We discuss the results of two norming studies (using regular English nouns as well as nonce nouns) which compare nonverbal nouns such as *ballerina*, for which we suggest they have an event argument, and nonverbal nouns such as *person*, for which we suggest they do not have an event argument. The results of the experiments point towards the fact that English speakers are sensitive to the noun type that a *skillful*-type adjective combines with, and, as a consequence, to the presence of event arguments in nouns. We connect this to broader psychological concepts capturing how humans identify and categorize entities belonging to different kinds (Prasada & Dillingham 2006), and argue that *skillful*-type adjectives are sensitive to activities that are so-called *k*-properties associated with nouns, which we represent linguistically by means of an event argument in the semantics of the respective nouns.

Keywords: *adjectives, nouns, event arguments, linguistic knowledge, world knowledge*

1 Introduction

In this paper, we address the question of which part of world knowledge is linguistically represented by taking a closer look at adjective-noun constructions such as *skillful dancer*, *skillful ballerina*, and *skillful person*. Specifically, we address the question to what extent a particular type of world knowledge about typical activities associated with these nouns motivates the assumption of an event argument in the semantics of these nouns, be they deverbal or nondeverbal.

Larson (1998) was first to propose that deverbal nouns like *dancer* have an event argument. While this may, as Larson (1998) argues, capture the ambiguity of adjective-noun constructions like *beautiful dancer* (‘beautiful and a ballerina’ vs. ‘beautiful as a ballerina’), this approach raises the question for which nouns an event argument can be assumed. An event argument is mostly assumed to be present in deverbal nouns only (Larson 1998, Winter & Zwarts 2013, Alexeyenko 2015, Maienborn 2020). Rapp (2015), on the other hand, assumes that deverbal as well as nondeverbal nouns can semantically be associated with events, together making up the class of eventive nouns. From a meaning-based perspective, this seems appropriate. In noun pairs like *dancer* – *ballerina* and *piano player* – *pianist*, it seems hard to justify why the deverbal noun should have an event argument, but not the nondeverbal noun. Importantly, both types of nouns can lead to a default reading when combining with adjectives like *skillful* and *talented*, which we assume are underspecified regarding what the respective skill/talent is. *Skillful dancer* and *talented ballerina* can be paraphrased as ‘skillful as a dancer’ and ‘talented as a ballerina’, respectively, which indicates that these nouns can resolve the underspecification of the adjective. However, not all nouns are capable of doing so: *skillful woman* and *gifted boy* cannot be paraphrased as ‘skillful as a woman’ and ‘gifted as a boy’. We therefore take the interaction of the noun and the adjective to serve as a ‘window’ into the noun meaning.

We argue that deverbal nouns such as *dancer* as well as certain nondeverbal individual nouns such as *ballerina* have an event argument as part of their semantics (Rapp 2015, Reinert 2024), since they are linked to typical activities. In this, they contrast with personal stative nouns such as *person* and *woman*. Our experimental results show that the deverbal and nondeverbal nouns in question pattern with each other in terms of acceptability in *as*-phrase constructions such as *skillful as dancer*, *skillful as a ballerina*, and *skillful as a person*. Our results thus support a meaning-based perspective on the semantics of nouns (opposed to a purely morphological view). We connect these findings to insights from psychological research, which distinguishes different types of properties associated with the concept behind the respective noun (Prasada & Dillingham 2006, Prasada & Dillingham 2009, and Prasada et al. 2013).

This paper is structured as follows. In Section 2, we provide an intersective analysis of adjective-noun constructions and discuss the relationship between nouns and events. Section 3 presents the results of a speeded acceptability judgment task testing the acceptability of *as*-phrases containing different noun types. In Section 4, we present the results of a small-scale offline acceptability judgment task testing *as*-phrases containing nonce nouns. Sections 5 and 6 discuss our findings and conclude the paper.

2 Theoretical background

2.1 An intersective analysis of adjective-noun constructions

We assume that adjectives like *skillful* and *talented* are underspecified: unless further specified or retrievable from contextual information, it is not clear which skill or talent is referred to. This becomes immediately clear in sentences such as (1).

- (1) Peter is skillful. (English)

The adjective's underspecification can be captured by a context-dependent parameter R in the adjectival semantics, which represents a relation between events and individuals (i.e. R is of type $\langle v\langle et \rangle \rangle$), corresponding to an activity the individual performs in the event.¹ Since they are gradable, these adjectives also have a context-dependent comparison class C as a parameter.

- (2) $\|skillful\| = \lambda x.[skillful_w(R)(C)(x)]$

We follow a pragmatic account for the default reading, i.e. the reading where the adjectival parameter is dependent on the noun that the adjective modifies. Being surrounding linguistic material to the adjective, the noun is generally accessible for the adjective as context. A pragmatic principle, adopted from Maienborn (2020), ensures that the information provided by the modified noun is generally given preference when determining the value of the adjectival parameter.

- (3) Preference principle for the pragmatic specification of free variables
(Maienborn 2020: 78, based on Maienborn 2003: 496):
Free variables are instantiated preferentially by linguistically introduced material, always provided that all the given requirements are met.

However, not every noun can deliver a comparison property for the interpretation of the adjective-noun construction. To be able to do so, the noun must allow the derivation of a typical activity.

- (4) Requirement for the use of the noun (Reinert 2024: 64):
The noun must allow to derive a typical or characteristic activity (which will then serve as the comparison property).

A default reading arises only for nouns that fulfill this requirement. The adjective phrase and the noun phrase (both of type $\langle et \rangle$ after the nominal event argument, if present, is bound by a generic quantifier) intersect. We propose that the ability of a noun to fulfill the above requirement is reflected in its semantics: since they may provide a default for the adjectival parameter, we assume that both *dancer* and *ballerina* have an event argument. When *skillful* combines with *dancer* or *ballerina*, R can therefore be identified with the noun, since they

¹For further theoretical background and discussion of this approach see Reinert (2024). We are very thankful to Ede Zimmermann for his support in developing the analysis presented in Reinert (2024) further, leading to the analysis adopted in this paper.

are of the same type $\langle v\langle et \rangle \rangle$. This is not possible when *skillful* combines with *woman*. We capture the observation that there is no default interpretation when nouns like *woman* and *person* combine with the adjective by assuming that they do not have an event argument. Since the noun therefore is of the wrong type, R cannot be identified with the noun in these cases, which is why (7b) is not even a well-formed formula. R must instead be found in the context in these cases.

- (5) a. $\|dancer\| = \lambda e.\lambda x.[dancer_w(e)(x)]$
 b. $\|dancer_{GEN}\| = \lambda x.[GENe\ dancer_w(e)(x)]$
 c. $\|skillful\ dancer\|$
 $= \lambda x.[skillful_w(R)(C)(x) \wedge GENe\ dancer_w(e)(x) \wedge R = dancer]$
 $= \lambda x.[skillful_w(dancer)(C)(x) \wedge GENe\ dancer_w(e)(x)]$
- (6) a. $\|ballerina\| = \lambda e.\lambda x.[ballerina_w(e)(x)]$
 b. $\|ballerina_{GEN}\| = \lambda x.[GENe\ ballerina_w(e)(x)]$
 c. $\|skillful\ ballerina\|$
 $= \lambda x.[skillful_w(R)(C)(x) \wedge GENe\ ballerina_w(e)(x) \wedge R = ballerina]$
 $= \lambda x.[skillful_w(ballerina)(C)(x) \wedge GENe\ ballerina_w(e)(x)]$
- (7) a. $\|woman\| = \lambda x.[woman_w(x)]$
 b. $\|skillful\ woman\|$
 $= \lambda x.[skillful_w(R)(C)(x) \wedge woman_w(x)]$
 $\neq \lambda x.[skillful_w(woman)(C)(x) \wedge woman_w(x)]$ (not even well-formed!)

2.2 Nouns and events

Given this approach, the relevance of the question whether the noun has an event or not becomes clear. As Larson (1998: 17) puts it, “how pervasive is event modification in nominals?” Under our approach, this presents itself as the question which nouns fulfill the requirement in (4). An initial answer to this question can follow from linguistic tests such as the one applied in Experiment 1 in Section 3: nouns that can appear in an *as*-phrase are able to provide a value for the adjectival parameter R , and thus are assumed have an event argument in our approach.

However, this does not answer the deeper question *why* certain nouns are able to provide a value for the adjectival parameter, while others are not. We propose that insights from psychology show the way forward with respect to this question: Prasada & Dillingham (2006), Prasada & Dillingham (2009), and Prasada et al. (2013) propose that concepts (expressed by nouns) are associated with different qualities of properties. According to them, *k*-properties are “properties of a token that are determined by the type of thing it is”, while *t*-properties are “properties [of a token] that are not determined by the type of thing it is”, they are merely statistical connections instead. To identify *k*- and *t*-properties, Prasada and colleagues propose, among other tests, the *by virtue of*-test (for English): Since (8) is acceptable, *four-legged* qualifies as a *k*-property of *dog*, and since (9) is not acceptable, *brown* does not qualify as a *k*-property of *dog*, but is rather classified as a *t*-property, since most dogs are brown.

- (8) Dogs, by virtue of being dogs, are four-legged.
- (9) #Dogs, by virtue of being dogs, are brown.

We suggest that this distinction between *k*- and *t*-properties is linguistically relevant for the decision whether a noun should have an event argument or not. We therefore postulate the following hypothesis:

- (10) Hypothesis:
Only activities that are *k*-properties are linguistically represented, i.e. only nouns with activities as *k*-properties have an event argument.

We assume that nouns like *dancer* and *ballerina* have an activity (or more activities) as a *k*-property, e.g. ‘dancing’. Importantly, there are also nondeverbal nouns representing concepts that are connected with activities that are *k*-properties. That is, ‘(ballet type) dancing’ is a *k*-property connected with the concept behind *ballerina*, since there *must* be some (ballet type) dancing in order for someone to be rightfully called *ballerina*. This is captured by assuming an event argument for nouns like *ballerina* as well, which consequently are able to interact with *R* in the exactly the same way as nouns like *dancer*.

Other nondeverbal nouns like *woman*, *person*, and *baby*, on the other hand, are not associated with activities as *k*-properties, but only as *t*-properties, e.g. ‘crying’ or ‘crawling’ in the case of *baby*. As a result, we propose that only activities that are *k*-properties are accessible to serve as the comparison property of adjectives like *skillful* in default readings of the adjective-noun constructions. In this sense, we suggest that the distinction between *k*- and *t*-properties can be viewed as reflecting the border between linguistic knowledge and world knowledge.

3 Experiment 1: Testing the acceptability of *as*-phrases

In this section, we present and discuss a speeded acceptability judgment task exploring the acceptability of English deverbal eventive, nondeverbal eventive, and personal stative nouns in *as*-phrases, building on the *as*-phrase test introduced above. To foreshadow our results, we find that both types of eventive nouns were rated as acceptable almost unanimously by our participants while personal statives were rated as significantly less acceptable in *as*-phrase constructions. Testing the *as*-phrase test more systematically therefore reveals that eventive nouns pattern alike in terms of their acceptability, irrespective of their deverbality.

3.1 Research questions and hypotheses

The first norming experiment is aimed at investigating if there is a difference between eventive nouns and personal statives in the first place. As discussed above, personal statives should not be able to deliver a default skill or activity when combined with *skillful*-type adjectives, while this should be possible with eventive nouns. If the *as*-phrase test is a valid test to distinguish these two noun classes, then their behavior can be tested by comparing the acceptability of these noun classes in *skillful as a ...*-phrases. An analysis that is in line with Larson (1998) would also predict a difference between deverbal and nondeverbal eventive

nouns in that acceptability should drop if a nondeverbal eventive noun is inserted into an *as*-phrase. In short, we hypothesize that the presence of an eventive noun should lead to a default skill being accessible, which should be reflected in the acceptability of an *as*-phrase containing an eventive noun.

We tested two different adjectives, namely *skillful* and *talented*. Compared to *skillful*, *talented* is a much more frequent adjective and occurs more often in combination with our nouns of interest in corpora such as enTenTen21 (via SketchEngine). We included both adjectives to exclude that frequency effects connected to the adjective used interfered with the acceptability of individual adjective-noun combinations tested. The hypotheses tested in the first norming experiment are therefore the following:

1. Eventive nouns and stative nouns differ in their ability to deliver a default skill that can serve as comparison property for a *skillful*-type adjective. Eventive nouns can deliver a default skill or activity, while stative nouns cannot do so.
2. There is no significant difference in acceptability when deverbal and nondeverbal eventive nouns are tested.
3. There is no significant difference in acceptability between different adjective-noun combinations involving the less frequent adjective *skillful* and the more frequent adjective *talented*.

3.2 Materials

We designed 36 experimental items such as the ones shown in (11):

- (11) a. Alice is skillful/talented as a dancer. (deverbal eventive)
 b. Julia is skillful/talented as a ballerina. (nondeverbal eventive)
 c. Max is skillful/talented as a person. (personal stative)

The items followed a 2x3 within-subjects design with the factors *Adjective* (*skillful* vs. *talented*) and *Noun class* (deverbal eventive, nondeverbal eventive, or personal stative). Participants saw all 36 nouns (12 from each of the three noun classes), but they never saw one and the same noun with both adjectives.

The critical items were accompanied by 36 filler items which contained *for*-phrases instead of *as*-phrases as well as different adjectives (e.g. *tall*, *old*, and *blonde*). Filler items were of varying acceptability, mirroring the differences in acceptability that we anticipated to show in our critical items.

3.3 Procedure

We decided to limit the amount of time that participants had for making their judgment to avoid coercive processes that were not of immediate interest. In theory, it is possible to meaningfully interpret phrases such as *skillful as a person*, but finding a meaningful interpretation here through coercive processes should take longer than accessing the default skill or activity, if present.

The experiment was conducted online on PClbex Farm (Zehr & Schwarz 2018). Subjects rated the items as acceptable or unacceptable by keying ‘f’ or ‘j’, respectively. They had 3.5 seconds to do so before timing out and moving on to the next trial. Before completing the experimental trials, participants gave informed consent, were introduced to the task at hand, and completed a few practice trials. The experiment lasted about 12 minutes.

We predict that *as*-phrases containing eventive nouns should be judged as acceptable by the participants. *As*-phrases containing personal statives should generally be unacceptable, with only few ‘f’-responses as reaction to them. There should be no difference in acceptability between deverbal and nondeverbal eventive nouns. We also do not expect adjective frequency to play a role for the acceptability of *as*-phrases, meaning that acceptability ratings for these phrases should not differ significantly if they are preceded by *skillful* compared to *talented*.

3.4 Participants

40 monolingual speakers of American English participated in the experiment. Subjects were recruited on Prolific Academic and completed the experiment on their own laptops or desktop computers. They were compensated for their time at a rate of \$12/hour.

3.5 Analysis

Prior to statistical analysis, we excluded four participants because they achieved less than 75% accuracy in their responses to the either clearly acceptable or unacceptable filler items, a threshold for exclusion that we set prior to collecting data. This left data from 36 subjects for analysis.

For statistical analysis, we adopted a frequentist approach and fitted a logistic mixed effects model with the fixed effects *adjective* and *noun class* as well as *item* and *participants* as random effects in R (R Core Team 2024). The model included random intercepts by participant and by item. *Adjective* was sum-coded, and *Noun class* was Helmert-coded. The levels *deverbal eventive* and *nondeverbal eventive* were first compared with each other, and then the grand mean of both was compared to *personal stative* because those are the two comparisons that we were interested in.

3.6 Results

Figure 1 shows the proportion of ‘f’-responses per adjective-noun combination in the different experimental conditions.

The figure clearly shows the difference in acceptability ratings between the eventive nouns in comparison to the personal statives. The difference between deverbal eventive nouns and nondeverbal eventive nouns is small, and items belonging to the three noun classes were rated similarly independent of the adjective that they were presented with (e.g. *skillful* or *talented*).

The output of the logistic mixed effects model is shown in Table 1. The comparison between the ratings given to *as*-phrases containing personal statives and the grand mean of the ratings of all items containing eventive nouns shows that the presence of a personal stative

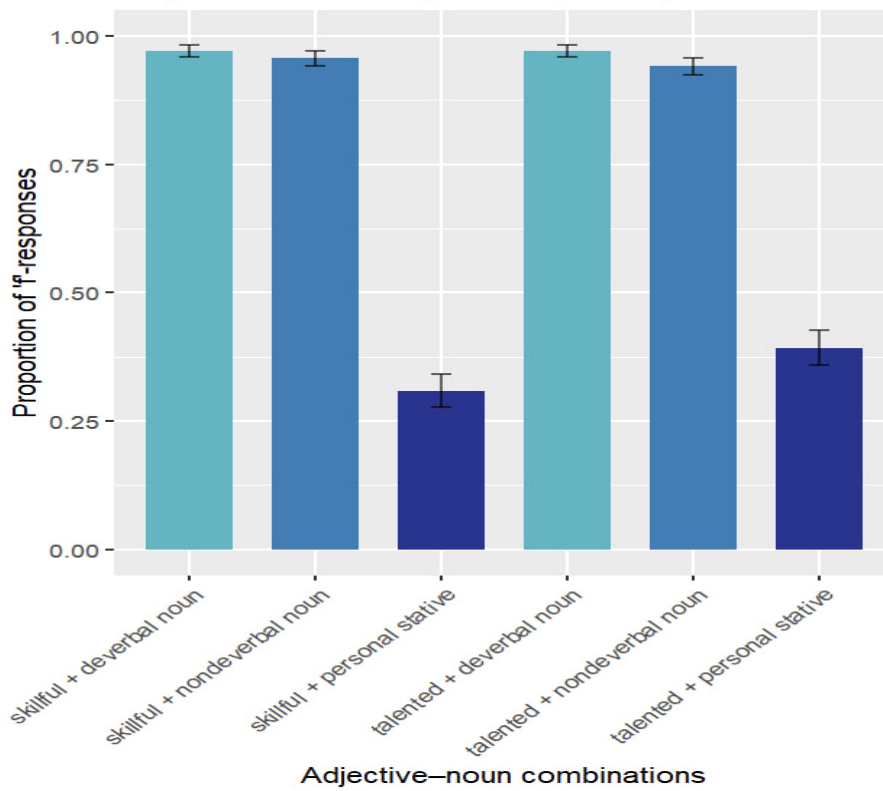


Figure 1 Proportion of 'f'-responses (indicating acceptability) per adjective-noun combination in Experiment 1. Error bars represent standard errors of the mean proportion.

significantly increases the number of 'j'-keypresses. No other significant main or interaction effects were found in the model, which is in line with our predictions.

Fixed effects	Estimate	Std. Error	z value	Pr(> z)	Signif.
Adjective	-0.06545	0.31685	-0.207	0.836	
Deverbal vs. Nondeverbal eventives	-0.62692	0.43617	-1.437	0.151	
Eventives vs. Personal statives	5.05596	0.35738	14.147	< 2e-16	***
Adjective x Eventives	-0.31328	0.87223	-0.359	0.719	
Adjective x Eventives vs. Personal statives	-0.71376	0.57747	-1.236	0.216	

Table 1 Experiment 1: Output of the logistic mixed effects model

3.7 Discussion

The results point towards the fact that there is a difference between eventive nouns and personal statives with respect to their ability to deliver a default skill or activity in *as*-phrases. This difference does not seem to be modulated by whether an eventive noun is deverbal or not, at least what the acceptability judgments investigated here is concerned. There was also

no significant difference between the two adjectives tested, ameliorating concerns that the frequency of the adjective tested could play a role here.

The results of Experiment 1 are consistent with the meaning-based account sketched out above relying on the connection of the noun with activities that are *k*-properties, and thus the presence or absence of an event argument in a noun. An explanation along these lines would predict that nouns that do not provide a default activity, such as personal statives, do not have an event argument in their semantics, making them unacceptable in the items tested here. The *as*-phrase test seems to be a reliable test to determine whether a noun is eventive or stative. However, there is a potential confound that this experiment did not address. The results are also consistent with an account based on differences in the structure of world knowledge between the different nouns tested. The idea here would be that humans have more specific world knowledge about what e.g. a ballerina does, compared to what a person does. This could also explain the difference in acceptability, as subjects could have had more targeted world knowledge available that allowed them to derive a default skill for a noun such as *ballerina*, making these nouns more acceptable than personal statives. We conducted a small-scale follow-up experiment that we report on in the next section to address parts of this confound.

4 Experiment 2: Telling apart world knowledge and semantic structure

As already discussed, there is a potential confounding variable in Experiment 1 that could have had an influence on the outcome, namely the extent of world knowledge associated with a noun. If the extent of world knowledge about activities associated with different kinds of nouns influences the acceptability of individual adjective-noun constructions, then the differences in acceptability found in the previous experiment cannot be based on different linguistic cues present in the nouns alone. Using nonce nouns instead of real nouns could help to tease apart the influence of world knowledge and linguistic cues on the acceptability of the adjective-noun constructions tested here. Once the specific world knowledge that is associated with nouns such as *ballerina* or *person* is taken away, and once the amount of world knowledge provided to participants is controlled for, we should be able to see more clearly whether the acceptability judgments are driven by world knowledge or by the linguistic cues contained within the nouns.

4.1 Research questions and hypotheses

Parallel to the hypotheses posed for Experiment 1, we formulate the following hypotheses for Experiment 2:

1. Eventive nonce nouns and stative nonce nouns differ in their ability to deliver a default skill that can serve as comparison property for a *skillful*-type adjective. Eventive nonce nouns can deliver a default skill or activity, while stative nonce nouns cannot do so.
2. As the results from Experiment 1 suggest, there is no difference in acceptability when different classes of eventive nonce nouns (e.g. deverbial and nondeverbial eventive nonce nouns) are tested.

4.2 Materials

The design of the nonce noun task closely orientates itself on the design of Experiment 1, except for certain aspects of it which need to be changed since the nonce nouns need to be learned and understood by the participants. As we did not find a significant difference between the ratings of items based on the adjective used, we abandoned the 2x3-design used in Experiment 1, turning Experiment 2 into an experiment with a 3x1 design. Experiment 2 was also not speeded, but a regular acceptability judgment task. There are two reasons for this. First, the subjects needed to be exposed to and briefly learn the new nonce nouns, and they needed more time for that as if they were judging the acceptability of real nouns. Second, we assumed that it would be harder for the participants to derive the undesirable coerced interpretations discussed above since the nonce nouns should not allow for coercion of this kind. This took away one main motivation for running a speeded acceptability judgment task.

In the materials, the nonce nouns had to be introduced in the context of the critical item so that the participants had a chance to understand what the nonce nouns could mean. (12) shows a sample set of (English) stimuli featuring nonce nouns:

- (12)
- a. The aliens on Zocollo like to do different things during their free time. Some aliens, such as Jerry, are *fraplers*. Jerry is skillful as a *frapler*. (deverbal eventive)
 - b. The aliens on Zocollo have different kinds of jobs. Some aliens, such as Lisa, are *sorlets*. Lisa is skillful as a *sorlet*. (nondeverbal eventive)
 - c. The aliens living on Zocollo belong to different species. Some aliens, such as Max, are *blikens*. Max is skillful as a *bliken*. (personal stative)

Deverbal nouns can be clearly recognized as such due to the fact that they always end in *-er*. Nondeverbal eventive nouns and personal statives cannot be distinguished by such morphological cues, so context is essential for the participants to determine which noun class a nonce noun belongs to. Therefore, all nondeverbal eventive items were preceded by a context that suggested that the nonce noun denoted some kind of profession, and the personal stative items were preceded by context that suggested that the nonce noun denoted some fixed characteristic of an individual, such as their species or family membership.

4.3 Procedure

When introduced to the task, participants were informed that they would be learning nouns from a language spoken by aliens on a fictive planet and would be asked to judge the acceptability of sentences containing the new words that they just learned.

Again, subjects rated the items as acceptable or unacceptable by keying 'f' or 'j', respectively. The results of Experiment 1 suggested that the contrast between eventive nouns and stative nouns is a rather large effect. Since we assumed it to be more difficult for the participants to judge the acceptability of nonce nouns compared to real nouns, we planned on making Experiment 2 shorter than Experiment 1. As a result, participants judged 12 critical items only (four items from each noun class condition). These experimental items were

interspersed with 12 filler items similar to those used in Experiment 1 that featured nonce nouns as well this time.

We predict that *as*-phrases containing eventive nonce nouns should be judged as acceptable by the participants, with only few ‘j’-responses as reaction to them. We do not predict a difference in acceptability between deverbal and nondeverbal eventive nonce nouns, meaning that both classes of nouns are rated as equally acceptable. *As*-phrases containing personal stative nonce nouns should be judged as unacceptable by the participants, with only few ‘f’-responses as reaction to them.

4.4 Participants

15 monolingual speakers of American English participated in Experiment 2. They were recruited and compensated as described for Experiment 1.

4.5 Analysis

All statistical procedures were identical to Experiment 1, with the only difference being the absence of *adjective* as fixed effect in Experiment 2. Prior to statistical analysis, we excluded two participants with the same exclusion criteria as applied in Experiment 1, leaving data from 13 subjects for analysis.

4.6 Results

The proportion of ‘f’-responses across the three experimental conditions is shown in Figure 2. Overall the response pattern found in Experiment 2 mirrors the one that we observed in Experiment 1. Items containing deverbal and nondeverbal eventive nonce nouns were rated as acceptable almost unanimously, and acceptability dropped in the case of personal statives, even though the drop is not as steep as in Experiment 1.

The logistic mixed effects model output is shown in Table 2. Again, the presence of a personal stative in an experimental item significantly increased the number of ‘j’-keypresses. We did not find other significant main effects.

Fixed effects	Estimate	Std. Error	z value	Pr(> z)	Signif.
Deverbal vs. Nondeverbal eventives	-1.4939	0.9554	-1.564	0.11791	
Eventives vs. Personal statives	3.3262	0.7632	4.358	1.31e-05	***

Table 2 Experiment 2: Output of the logistic mixed effects model

4.7 Discussion

Overall, we replicated the results of Experiment 1 using nonce nouns in Experiment 2. Again, our prediction that deverbal and nondeverbal eventive nouns are able to deliver default skills equally well is borne out. Our prediction that personal statives should be rated as less acceptable is borne out as well. However, items containing personal statives were rated as

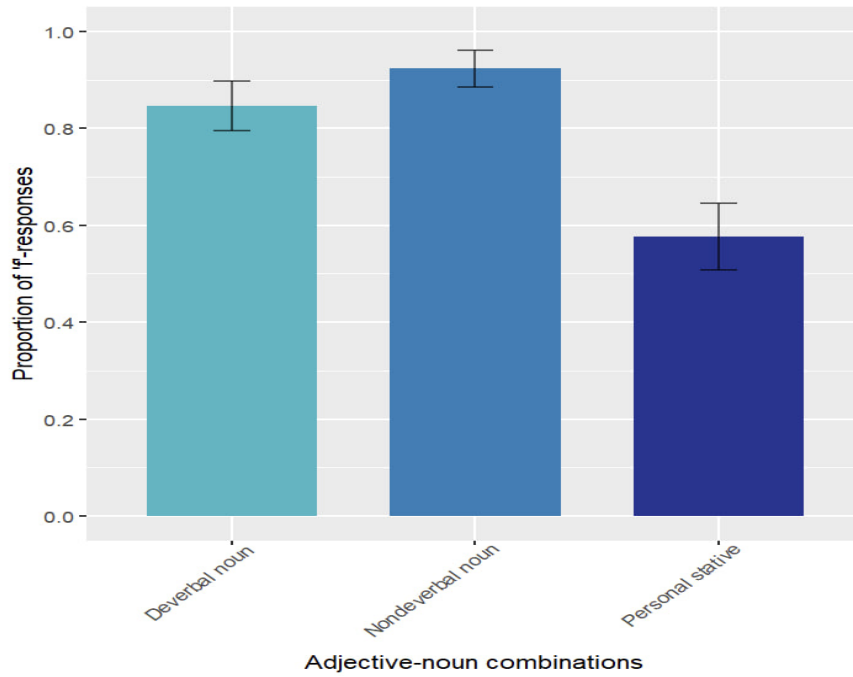


Figure 2 Proportion of ‘f’-responses (indicating acceptability) per adjective-noun combination in Experiment 2. Error bars represent standard errors of the mean proportion.

acceptable more often than in Experiment 1 in our nonce noun experiment. This could have several reasons. One factor could be the extent to which individuals took the context sentences preceding the nonce nouns into account when making their judgments. Apart from that, our sample size was much smaller in Experiment 2, making the results of both experiments not directly comparable in that respect.

Crucially, we cannot rule out the influence of world knowledge on acceptability judgments entirely (if that is even possible). Teaching participants the meaning of nonce nouns requires introducing the nouns in contexts that are prototypically associated with either eventive nouns or personal statives. It is therefore still possible to associate the broader class of eventive nonce nouns with activities, even though subjects cannot determine the exact activity associated with a nonce noun. The results are therefore still consistent with an account based on world knowledge instead of the semantic account discussed above that postulates an event argument in the semantics of eventive nouns. However, the design introduced here is able to rule out the influence of world knowledge associated with individual nouns tested here, which is a step in the right direction.

5 General discussion

The results from both experiments conducted here point in the same direction. Whether a noun was deverbal or not did not seem to play a role when participants were asked to judge the acceptability of deverbal and nondeverbal nouns appearing in *as*-phrases. While deverbal as well as nondeverbal eventive nouns were constantly rated as highly acceptable,

items containing personal statives were being given significantly lower acceptability ratings. This pattern was visible not only with regular English nouns, but also with the nonce nouns that we tested in Experiment 2. Testing nonce nouns instead of real nouns allowed us to partially deconfound the influence of world knowledge associated with individual nouns that speakers have and the predictions that our meaning-based account makes. While it is not possible to fully tease apart world knowledge and semantic structure in Experiment 2, as discussed above, the results of Experiment 2 alleviate some of our worries regarding the influence of the extent of associated world knowledge. The results make it less likely that more specific knowledge e.g. about what dancers in general do influenced participants to assign higher ratings to an item containing *skillful as a dancer*, compared to items such as *skillful as a person*, which would be expected to come with less specific world knowledge regarding what a person in general does.

Returning to the theoretical questions posed above, we conclude that our results are compatible with a meaning-based account that poses the presence of an event argument in the structure of all eventive nouns tested here, irrespective of whether they are deverbal or not. Our results clearly show that all of these nouns can provide a prototypical activity, which we capture as an event argument in their structure. The distinction between the two classes of eventive nouns on the one hand, and personal statives on the other hand, also naturally falls out when connecting our results back to the distinction between *k*- and *t*-properties as discussed in Section 2 (Prasada & Dillingham 2006, Prasada & Dillingham 2009 and Prasada et al. 2013). While *dancer*- and *ballerina*-like nouns have a linguistically represented *k*-property that speakers can resort to when accessing the comparison property of adjectives such as *skillful*, *person*-like nouns do not have activities as *k*-properties that could be used as the adjective's comparison property. Distinguishing *k*- and *t*-properties, as well as connecting them to the presence or absence of an event argument in a noun's semantic structure, therefore suggests how the cut between semantic knowledge and world knowledge could be made in this domain.

6 Concluding remarks

In this paper, we introduced two experiments aiming at understanding how English speakers determine whether a noun is associated with a prototypical activity that can be used as input to the comparison property parameter present in *skillful*-type adjectives, and what kinds of linguistic and nonlinguistic cues help them with this task. We gave an overview over the semantics of *skillful*-type adjectives and different ways to classify individual nouns in Section 2 and moved on to the presentation of a norming experiment in Section 3. The follow-up experiment presented in Section 4 aimed at addressing the influence of the extent of associated world knowledge as confounding variable in Experiment 1. As discussed in Section 5, the effect of world knowledge on the perception of whole noun classes cannot be fully ruled out with the design used for Experiment 2. However, the results of the nonce noun experiment could help with ruling out or diminishing the effect of world knowledge associated with individual nouns that we tested in Experiment 1. A world knowledge-based account of the differences between noun classes is still possible. Experimental designs that help to decide completely in favor of or against the meaning-based approach that we developed here

could be explored in future work. Irrespective of that, the experiments presented here are a first step towards understanding how world knowledge and linguistic cues interact in the derivation of prototypical activities from nouns.

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